Inhibitor Investigation

One-page teacher/presenter guide · Murray State University · Bryant Harrison

Purpose

Students use a four-measurement virtual assay bench to distinguish among three finite, declared *synthetic initial-rate patterns*: competitive, uncompetitive, and pure noncompetitive. They learn that a point can be exact and still be insufficient evidence.

> **Model.** `v = Vmax·S / (a·Km + b·S)`, with Vmax = 100, Km = 2, and inhibition factor 3. Baseline uses a=b=1; competitive a=3,b=1; uncompetitive a=1,b=3; pure noncompetitive a=b=3.

First-load / presenter flow

1. Open `index.html` locally; the **Guide** overlay opens automatically. Use **?** to reopen it at any point.
2. Ask students which assay would be informative before they click. They may select 0, 1, 2, 4, 8, or 20 teaching units.
3. At S=Km=2, competitive and uncompetitive both predict 25.0 while pure noncompetitive predicts 16.7. Name this as an evidence boundary, not a failure.
4. For this exact model, S=1, 8, or 20 each eliminates a candidate after that tie. Do not rank one as “best” by its curve gap; the model has no measurement-error or resolution rule.
5. Students must explain before the mystery pattern is revealed.
6. ⚙ **Settings** offers **Open Presenter Notes** (this guide, on screen), **Presentation mode** (a stripped, projector-sized chart), and **Reset** (a fresh mystery sample with the bench, conclusion, and transfer feedback cleared). Reset leaves Presentation mode as you set it.

Discussion prompts

- What claim is supported by the plotted points? What claim is still too strong?
- Why does repeating the same concentration not add discrimination?
- Which assumptions would a real assay need to test before treating a fitted pattern as mechanism?

Expected synthetic predictions

Substrate S	Competitive	Uncompetitive	Pure noncompetitive
0	0.0	0.0	0.0
2 (=Km)	25.0	25.0	16.7
8	57.1	30.8	26.7
20	76.9	32.3	30.3

Scope and scientific caveats

This is an exact, noiseless, ideal initial-rate activity at fixed conditions. The tie at S=Km is deliberately constructed by the shared factor-3 set; it is not a general competitive/uncompetitive diagnostic. It is not an inhibitor fit, drug screen, or molecular-binding assignment.

The three patterns do not represent mixed, partial, allosteric, tight-binding, time-dependent, irreversible, multiple-substrate, or non-steady-state behavior. A real investigation would test a broader substrate/inhibitor matrix, controls, replicates, assay linearity, error, and model assumptions.

The activity intentionally lets students conclude **insufficient evidence**. It does not reward guessing the hidden generator, and it never claims that a curve pattern uniquely identifies a real mechanism.

Sources

Strelow J, Dewe W, Iversen PW, et al. Mechanism of Action Assays for Enzymes, NIH Assay Guidance Manual. Used for the inhibition-pattern and initial-rate assay context, including the value of measurements above Km.

A Murray State biochemistry course served as the reviewed course-fit reference. It is not named here, and it does not endorse this activity or supply current scheduling information.

Offline at runtime: no external scripts, styles, fonts, images, data, or AI service. Links are optional reader-initiated source citations.